Do Now · retrieval

Paging, segmentation and virtual memory

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is RAM used for?

Q2

What are remote virtual storage files?

Q3

Why might a process have to wait on memory I/O?

Q4

A 4-row, 3-column zero-indexed array: give its final valid index pair and number of values.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — Pages and frames have equal fixed sizes; the page table maps virtual pages to physical frames.

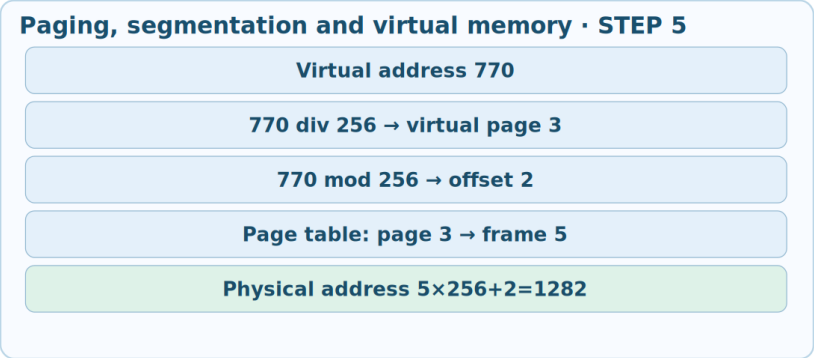
Core idea 2

Trace — Divide the virtual address by page size for the page number; use remainder as the offset.

Core idea 3

Explain — A page fault requires a non-resident page to be loaded. Variable-sized segmentation has different fragmentation characteristics.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

page — Virtual fixed-size memory block

frame — Physical fixed-size RAM block

page fault — Page absent from RAM

segmentation — Logical variable-length regions

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

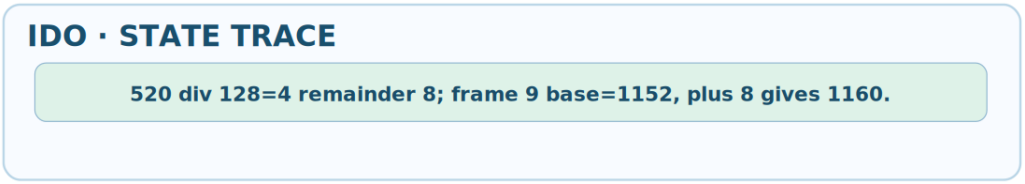
Problem

Translate virtual address 520 with page size 128 and page 4 mapped to frame 9.

Follow the worked method

1. Predict first — 520 div 128=4 remainder 8; frame 9 base=1152, plus 8 gives 1160.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

615 div 200=3 remainder 15. Page 3 maps to frame 7, hence physical address 1415.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Define page, frame, page table and page fault. [4]

Question 2

Page size 256, VA 770, page 3 mapped to frame 5. Find physical address. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Compare paging and segmentation and explain thrashing. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Paging changes the offset within the page when it changes frame.

Use key vocabulary from memory

1. Explain page in your own words.

2. Explain frame in your own words.

3. Explain page fault in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.